SIADS 696 Milestone II
Healthcare Access Characteristics: Predicting Routine Checkup Rates and Identifying Meaningful Groups by County

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Overview:
This project will examine how **healthcare-resource availability and community vulnerability relate to preventative healthcare utilization across U.S. counties**.

For the **supervised** portion, the main goal is to determine **how well county-level healthcare access characteristics** – such as primary-care provider supply, healthcare infrastructure , provider-shortage designations, and rurality – **can predict the percentage of adults receiving a routine checkup within the past year**.

For the **unsupervised** portion, the goal is to **identify distinct types of U.S. counties based on combinations of healthcare resources and social vulnerability**. These clusters may reveal different forms of healthcare-access disadvantage that would not be captured by a single access or vulnerability score such as provider-to-population ratio or a composite SVI score.

A broader policy objective is to better understand whether communities with similar preventative-care utilization may face different underlying barriers, which could have implications for healthcare planning and resource allocation.

Challenges/Limitations:
- We will be joining multiple datasets from Health Resources and Services Administration (HRSA) and Centers for Disease Control and Prevention (CDC) . Geographic units, year coverage, variable definitions, and missing-data patterns may differ, so the datasets will need to be harmonized using county Federal Information Processing Standard (FIPS) codes and the closest reasonable time periods.
- Additionally, Area Health Resources Files (AHRF) contain thousands of potential variables. The team will select a manageable and theoretically justified subset of healthcare resource variables.
- CDC PLACES outcomes are **model-based estimates rather than direct measurements for every county**. PLACES uses Behavioral Risk Factor Surveillance System (BRFSS), Census population data, American Community Survey (ACS) data, and variables including age, sex race/ethnicity education, and county poverty in its small-area estimation procedure. To avoid data leakage, we will avoid using those same demographic variables as supervised predictors.
- To mitigate that issue, the primary supervised models will emphasize **AHRF and Health Professional Shortage Area (HPSA) healthcare-resource features** and avoid demographic variables directly used in PLACES target construction. Furthermore, county-level results describe communities rather than individuals, so conclusions will not be interpreted as individual-level causal relationships.

Existing/Related Work:
Bauer et al. (2022), "County-Level Social Vulnerability and Breast, Cervical, and Colorectal Cancer Screening Rates in the US, 2018."

The study examined **3,141 U.S. counties** and combined CDC PLACES with CDC Social Vulnerability Index (SVI) data to investigate whether social vulnerability was associated with county-level breast, cervical, and colorectal cancer screening rates. The researchers also considered urban/rural status, health insurance coverage, and primary-care physician availability. They found that counties with greater social vulnerability generally had lower cancer-screening rates.

Differences and Improvements:
- The outcome will be routine preventative checkup utilization rather than disease-specific cancer screening.
- We will incorporate a broader representation of healthcare access by integrating AHRF healthcare workforce/infrastructure variables and HRSA HPSA shortage measures, rather than primarily evaluation SVI.

- The project will also include a separate unsupervised analysis to identify county healthcare-access/vulnerability archetypes.
- Instead of a single association/regression framework, the project will compare at least three diverse supervised machine-learning model families, consistent with the eventual Milestone II report requirement.
- We will explicitly investigate the relative contribution of healthcare resources versus community barriers through feature importance and feature-group ablation.

Sam previously completed a California-focused healthcare-access analysis involving AHRF, HPSA, and SVI data. Gina's Milestone I project focused exclusively on SVI data as it related to diabetes outcomes. Sara's Milestone I project utilized CDC and Census data to investigate chronic disease indicators and comorbidities on the state-level. However, this Milestone II project can be treated as a new national project with new research questions and newly developed supervised and unsupervised ML analyses. Code from previous projects will not be reused.

Part A: Supervised Learning

Proposed Datasets:
The proposed supervised dataset will be a new, county-level analytical dataset constructed from multiple public sources.

Outcome — CDC PLACES https://www.cdc.gov/places/measure-definitions/prevention.html?utm_source
The proposed target is the county-level **age-adjusted prevalence of adults reporting a routine checkup within the past year.** CDC defines this as adults reporting that they visited a doctor for a routine checkup, such as a general physical exam, during the previous year. Both crude and age-adjusted prevalence estimates are available.

Primary predictors — HRSA Area Health Resources Files (AHRF)
https://data.hrsa.gov/topics/health-workforce/nchwa/ahrf?utm_source
AHRF contains county-level healthcare workforce and resource information. The current AHRF release is annual, and individual variables may represent different underlying years depending on source availability.

Healthcare-shortage predictors — HRSA HPSA
https://data.hrsa.gov/topics/health-workforce/shortage-areas/dashboard?utm_source
HPSA data identify geographic areas, populations, and facilities experiencing shortages of primary care, dental, or mental-health providers. HRSA's downloadable data are publicly available and refreshed frequently.

Planned Supervised Learning Approaches:
- **Elastic Net regression** will serve as an interpretable linear baseline. Regularization is useful because healthcare-resource variables may be correlated, and Elastic Net can reduce the influence of redundant predictors.
- **Support Vector Regression (SVR)** will provide a fundamentally different learning framework capable of modeling nonlinear relationships through kernel functions.
- **Gradient Boosting**—or Random Forest —will represent tree-based ensemble learning and can capture nonlinear relationships, thresholds, and interactions among healthcare-access characteristics.

Feature representations:
The primary representation should emphasize approximately **15–30 "theoretically justified" county-level variables**, rather than feeding thousands of AHRF variables into the models. Potential feature groups include:
- **Provider supply**: primary-care physicians, family physicians, nurse practitioners, physician assistants, mental-health providers, dentists.
- **Healthcare infrastructure**: hospitals, hospital beds, health centers, and other relevant facility measures.
- **Provider shortage**: primary-care HPSA status, HPSA score, estimated provider shortage/FTE shortage, and rural status where appropriate.
- **Geographic/access context**: rurality, population density, and similar structural access measures.
- **Selected social/access barriers**: potentially uninsured population, transportation barriers, housing burden, disability, or unemployment, subject to final leakage review.

External datasets or tools:

The most important supplemental dataset is the **CDC/ATSDR Social Vulnerability Index (SVI)**. The 2022 national SVI provides county-level measures and rankings covering 16 social factors organized into four themes. SVI will be more important for Part B than Part A. Any SVI variables added to supervised modeling will first be checked against PLACES' estimation inputs.

Possible technical tools include Python with Pandas for integration and cleaning, Scikit-learn for preprocessing/modeling/evaluation, and Matplotlib, Altair, or similar tools for visualization. Geospatial visualization may use GeoPandas if county maps are incorporated.

Evaluation and Visualization Methods:

- **Mean Absolute Error (MAE) -** Primary metric; expresses average prediction error in percentage points and is easily interpretable.
- **Root Mean Square Error (RMSE)** - Penalizes large errors more than MAE; used alongside MAE to help identify if there are subsets of counties driving poor model performance.
- **5-fold cross validation** - Used for all model performance comparisons
    - Elastic Net: regularization path and l1 ratio
    - SVR: Hyperparameter search results (C, epsilon, gamma)
    - Gradient Boosting: Train vs validation loss curves to check for overfitting
- **Predicted vs observed routine check-up prevalence -** One scatter plot per model to visualize calibration and identify ranges where predictions differ from observed values.
- **Residual distribution boxplots -** Side-by-side comparison of residual distributions across models to assess error spread and skew.
- **Permutation feature-importance plots -** Produces a comparable importance ranking when used consistently across all three models.
- **Feature-group ablation -** Compares changes in performance metrics when dropping completely unrelated feature groups to evaluate importance of a construct (income, education, access, etc.) instead of a single feature.
- **SHAP plots -** Visualize feature importance and direction while also allowing for comparison between model types on a common scale.

Part B: Unsupervised Learning

Proposed Datasets:

The unsupervised analysis will use the same harmonized county-level dataset built for Part A, but remove the PLACES outcome variable itself (county-level age-adjusted prevalence of adults reporting a routine checkup within the past year) to preserve independence between our supervised target and unsupervised structures. The unsupervised portion will also use the CDC/ATSDR Social Vulnerability Index (SVI) which provides county-level measures and rankings covering 16 social factors organized into four subthemes: socioeconomic status, household composition/disability, minority status/language, and housing/transportation.

Planned Unsupervised Learning Approaches:

- **K-means** will serve as a simple, easily interpretable starting point despite its simplifying assumptions. Its more uniform clusters may not hold for county-level resource/vulnerability data.
- **Hierarchical (agglomerative) clustering** will allow exploration of more complex and irregular cluster structures without selecting k upfront.
- **Gaussian Mixture Models** will provide the option of softer cluster assignments (i.e. allow counties to take on more than one archetype) which may provide more nuanced and overlapping boundaries for resource/vulnerability data.
- **DBSCAN** would be a possible exploratory step to flag outlier counties whose resource/vulnerability profiles don’t fit well into the primary cluster structure.

Feature representations:

We will maintain the same feature groups as Part A (provider supply, infrastructure, provider shortage, geographic context) in addition to the four SVI subtheme scores. As stated above, we will exclude the reported checkup-rate outcome (to avoid leakage) and likely exclude PLACES demographic variables (to avoid overlap and redundancy) using SVI scores instead.

Evaluation and Visualization Methods:

- **Silhouette score and/or Davies-Bouldin index** – Determine how well-separated archetypes actually are or if there is a lot of overlap.
- **PCA and t-SNE –** 2D projections of the county feature space, colored by cluster assignment, to visualize whether clusters are well-separated.
- **Choropleth maps** – U.S. counties colored by cluster assignment.
- **Radar/spider charts -** Plot clusters (overlay depending on final number of clusters) on a spider chart along multiple dimensions (e.g. provider supply, infrastructure, social vulnerability, rurality) to visually characterize archetypes.
- **Boxplots or violin plots –** Directly compare PLACES checkup-rate distributions by cluster to determine if similar-utilization counties have different archetypes.

Timeline and Team Contributions

Task	Timeline	Lead Team Member	Supporting Member(s)
GitHub Setup	Week 3	Sara	Sam, Gina
Data Documentation	Week 3	Sam	Gina, Sara
Data Cleaning and Processing	Week 4-5	Gina	Sara, Sam
Feature Engineering	Week 4-5	Sara	Gina, Sam
Exploratory Data Analysis	Week 4-5	Sam	Sara, Gina
Supervised Analysis	Week 5-6	Gina	Sam, Sara
Unsupervised Analysis	Week 5-6	Sam	Gina, Sara
Visualizations	Week 6	Sara	Sam, Gina
Report	Week 6-7	All	All